

COMP3314 Image Classification Challenge Report

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1 Dataset Analysis

The dataset contains 50000 training samples and 10000 testing samples. Each sample is a 32×32 color image, and is labelled by an integer in the range $[0, 9]$. The classes corresponding to the labels are shown in Table 1, with visualized examples.

Label / Class	Example	Label / Class	Example
0 - Airplane		5 - Dog	
1 - Automobile		6 - Frog	
2 - Bird		7 - Horse	
3 - Cat		8 - Ship	
4 - Deer		9 - Truck	

Table 1: Label-Class Correspondence with Examples

The sample count for the training dataset is shown in Figure 1. It can be observed that the imbalance of data among different classes is small enough to be omitted.

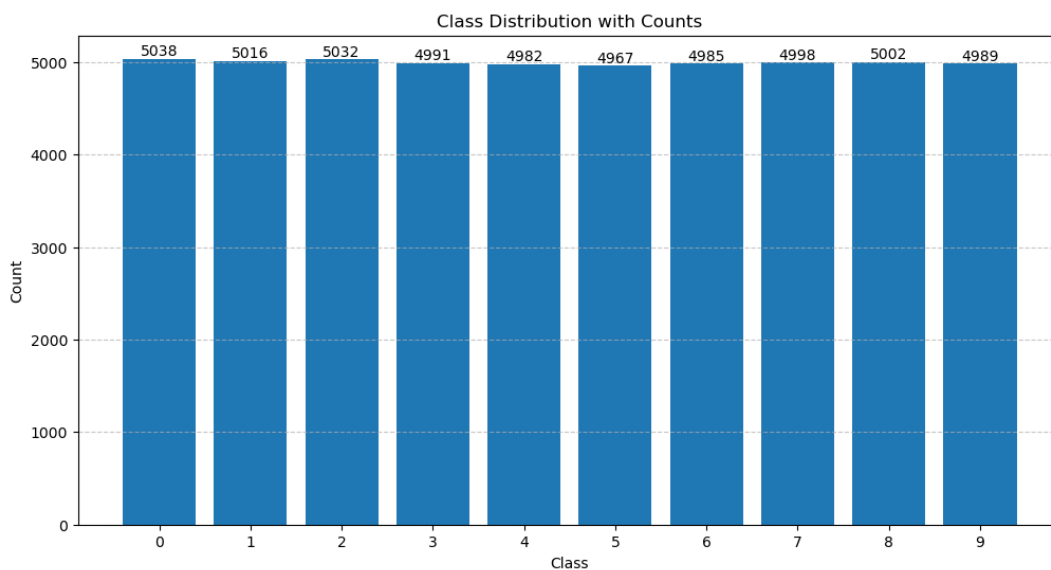


Figure 1: Training Set Samples by Class

2 Classifier Exploration

2.1 Performance Comparison

The data were preprocessed and features were extracted with various methods. Several classifiers were then trained on the processed data. After hyperparameter tuning, SVM could achieve an accuracy score of almost 75%, while Logistic Regression (LR), KNN and Random Forest (RF) showed poorer performances, with none gaining an accuracy of over 65%. A visualized comparison of the models' performances may be found in Figure 2. The confusion matrices for these 4 models may be found in Figure 3.

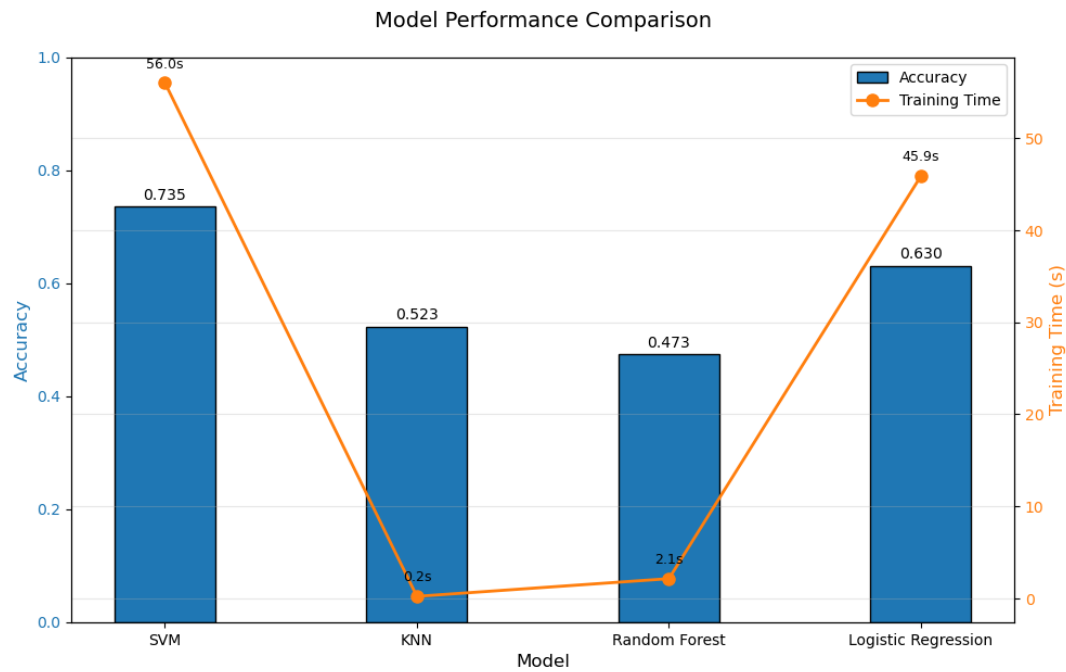


Figure 2: Performance Comparison between Models

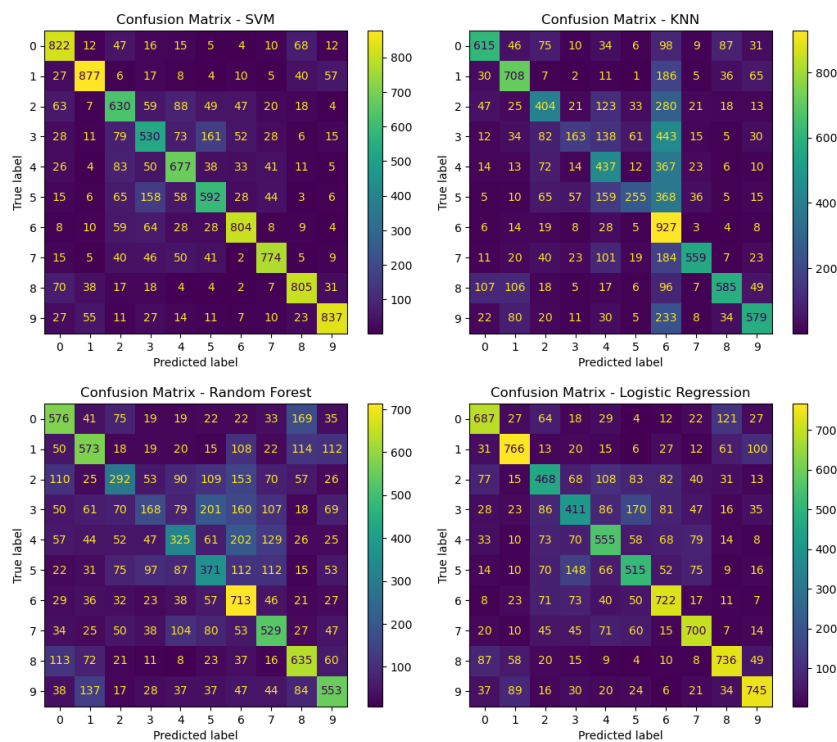


Figure 3: Confusion Matrices for Models

2.2 Analysis

SVM might have excelled from its high performance in high-dimensional feature spaces. The kernel trick helps find optimal decision boundaries from non-linearly separable data.

On the other hand, LR probably could not fit an accurate enough linear boundary for separating different classes. KNN suffers from the “curse of dimensionality”, where distance metrics tend to prove less helpful in higher dimensional spaces. The random splitting of RF could have led to omission of critical features. RF trees also tend to overfit more easily to noisy data.

3 Final Solution Description

Data read: The images are read and stored in RGB.

Feature extraction: Features extracted are DAISY descriptors, enhanced HOG, and HSV histograms. The features are standardized before concatenation.

Model training: An SVM model with an RBF kernel and parameter C set to 10 is trained.